

Response to the referee

We thank the reviewer for the careful reading of the revised manuscript and for the additional suggestions. We addressed all points raised. The manuscript and supplementary materials were updated accordingly.

Here we provide detailed response to each concern raised:

First general observation:

"The first is the method used to correct the earlier shortfall regarding whether the greater proportion of people in proximity to strict PAs is less than those in proximity to non-strict PAs is due to the number of PAs. The authors have looked at the areas of each category and then the number of people in proximity but I think a more effective way would be to look at the PA boundary distance instead of area."

This general concern is related to the following specific comment and they have been addressed jointly:

"Pg 5 Ln 16-22: I appreciate that the authors have addressed this issue as it makes a substantial difference to understanding the situation. However, I think the more appropriate way to measure this would have been comparing the total boundary lengths of strict vs non-strict PAs instead of the total area as, for example, many small non-strict reserves might be the same total size (in area) as a single large strict reserve, but the total boundary length would be much greater, affording increased possibility for people to live within 10km buffer (almost a reverse edge-effect!). If this is something that could be accomplished without too much difficulty – which I suspect it is given that all sites must have boundary lengths as attributes - it might be worth adding to the analysis."

We agree that total boundary length is a relevant complementary diagnostic for proximity-based exposure. We therefore extended the analysis to compute total PA boundary length by IUCN category from the same filtered WDPA geometries used in the manuscript. The new results confirm the previous interpretation: non-strict PAs remain more populated than strict PAs after standardization by total boundary length as well as by area.

In the revised manuscript, we now report both diagnostics. For the 2020 all-PA cross-section, non-strict PAs are associated with about 999 people inside or within 10 km per km of PA boundary, compared with about 891 for strict PAs. We also added a short methods note in the Supplementary Materials explaining how this boundary-length diagnostic was computed. The manuscript text was revised to state that the conclusion is unchanged under both approaches.

Second general observation:

"Secondly, I see a clear systematic bias with the comparison of the WorldPop vs GHSL estimates with ALL WorldPop estimates for people inside PAs being higher than their GHSL equivalents. Yet this is not mentioned. I think this needs to at least be mentioned and addressed in the discussion – even if the underlying reasons are unclear. I have detailed these more below."

We treated this comment together with:

"Figure 3: I'm not sure how this shows "broad agreement" at the Inside PA level as ALL of the estimates for this are above the line (they do not "cluster around the 1:1 line"), indicating that the WorldPop estimate is systematically higher than the GHSL estimate. This is reflected in Pg 11 Ln 8-9 but needs to be clearly stated within the figure as well. And since it is a systematic issue, it should be discussed (ideally on Pg 13 where these datasets are compared)."

We agree and revised both the Figure 3 treatment and the surrounding text. The previous wording was too broad. In the revised version, the Results section now distinguishes clearly between the two panels: agreement is described as relatively close for the broader "inside or within 10 km" perimeter, whereas the "inside PAs" panel is explicitly described as showing a systematic upward shift of WorldPop relative to GHSL.

We also modified Figure 3 itself by adding panel-specific annotations that summarize the bias pattern numerically. In the revised figure, the inside-PA panel reports that 64 of 75 country estimates lie above the 1:1 line, with a median WorldPop-to-GHSL ratio of 1.87. By contrast, the broader perimeter is much closer to parity, with a median ratio of 1.01.

The Discussion section now addresses this systematic asymmetry explicitly. We note that we cannot determine the precise cause from our data alone, but that the pattern is consistent with differences in how the two gridded products spatially allocate very small populations within PA interiors.

"Pg 5 Ln 7-8: It would be worth putting the % of population for those inside PAs (18.8 million) in 2020, as has been done for the 257.6 million inside or within 10km (15.4%)."

Done. We added the percentage corresponding to the 2000 inside-PA population in the Results section.

"Pg 6 Ln 12: Again, put the % here as it helps the reader to quickly understand the magnitude of the missing data."

Done. We now report the percentage of PAs with missing designation year alongside the count. In the revised version this is 59.4% of the PA records retained for the 75 LMIC sample.

"Pg 12 Ln 14: Combine sentences into one. "...Fisher et al. (2024) – yet they define...""

Done. The sentence in the Discussion section was rewritten as a single sentence.